

Coordinate-Independent Robot Model Identification

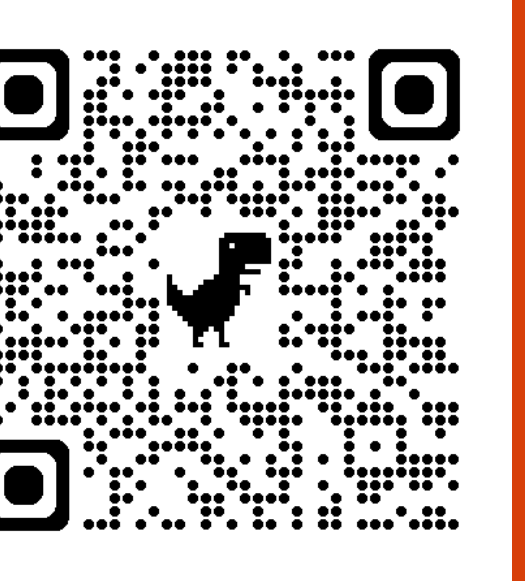
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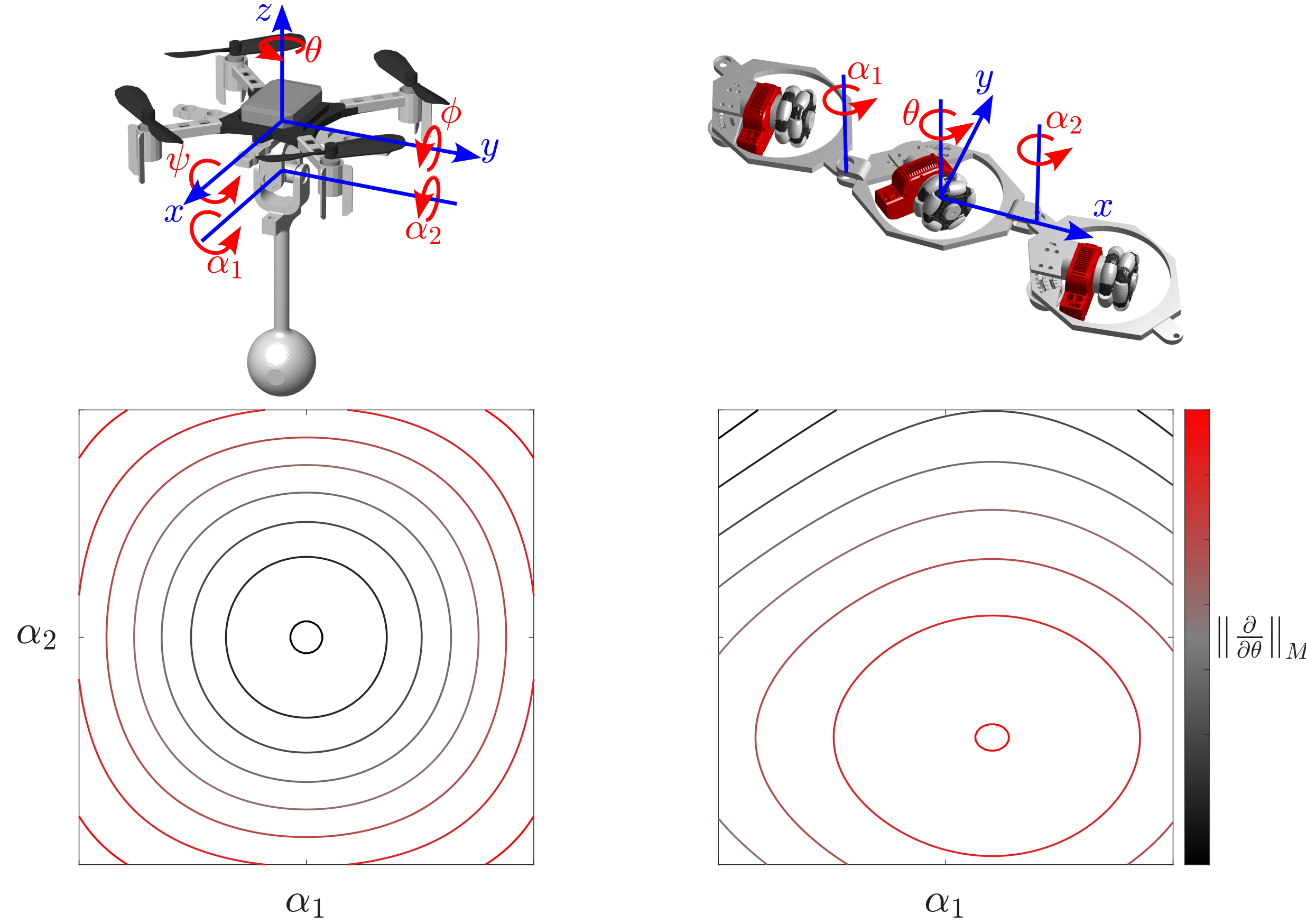


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01 Why coordinates matter

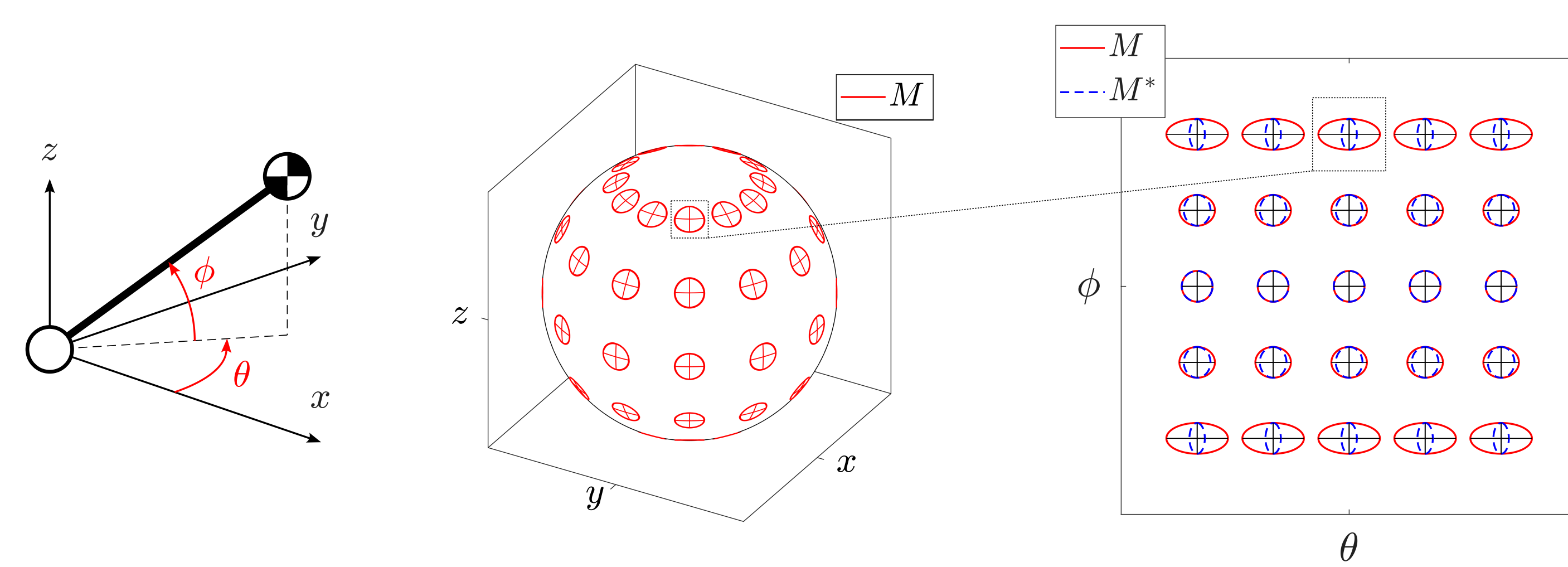
System identification often fits forward or inverse dynamics using Euclidean norms in generalized coordinates. But the same coordinate velocity or force can have different physical significance across configurations, units, and coordinate choices.



Same coordinate quantity, different physical significance. The same coordinate angular velocity can correspond to different kinetic energy (left) or drag dissipation (right) across (α_1, α_2) .

02 Fit residuals in the system's geometry

Define the inverse-dynamics residual $r(\pi) = Y\pi - \tau$. Because forces and velocities are dual through power, the velocity metric M induces the force-space metric $M^* = M^{-1}$.



Example: pan-tilt geometry. The ambient kinetic-energy geometry of the end-effector point mass pulls back to coordinate space as a velocity metric and its corresponding dual metric on force space.

Coordinate-independent system identification while preserving convexity

$$\min_{\pi} \sum_i r_i^T(\pi) M_i^{-1}(\pi) r_i(\pi) \iff \min_{\pi, \{s_i\}} \sum_i s_i$$

$$\text{s.t. } \begin{bmatrix} M_i(\pi) & r_i(\pi) \\ r_i^T(\pi) & s_i \end{bmatrix} \succeq 0.$$

The dual metric normalizes force residuals by physical significance.

For affine $M_i(\pi)$ and $r_i(\pi)$, the Schur complement gives an exact convex formulation. Physical-consistency constraints and geometric prior regularization remain compatible.

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03 Inertia-dominated: Crazyflie–pendulum

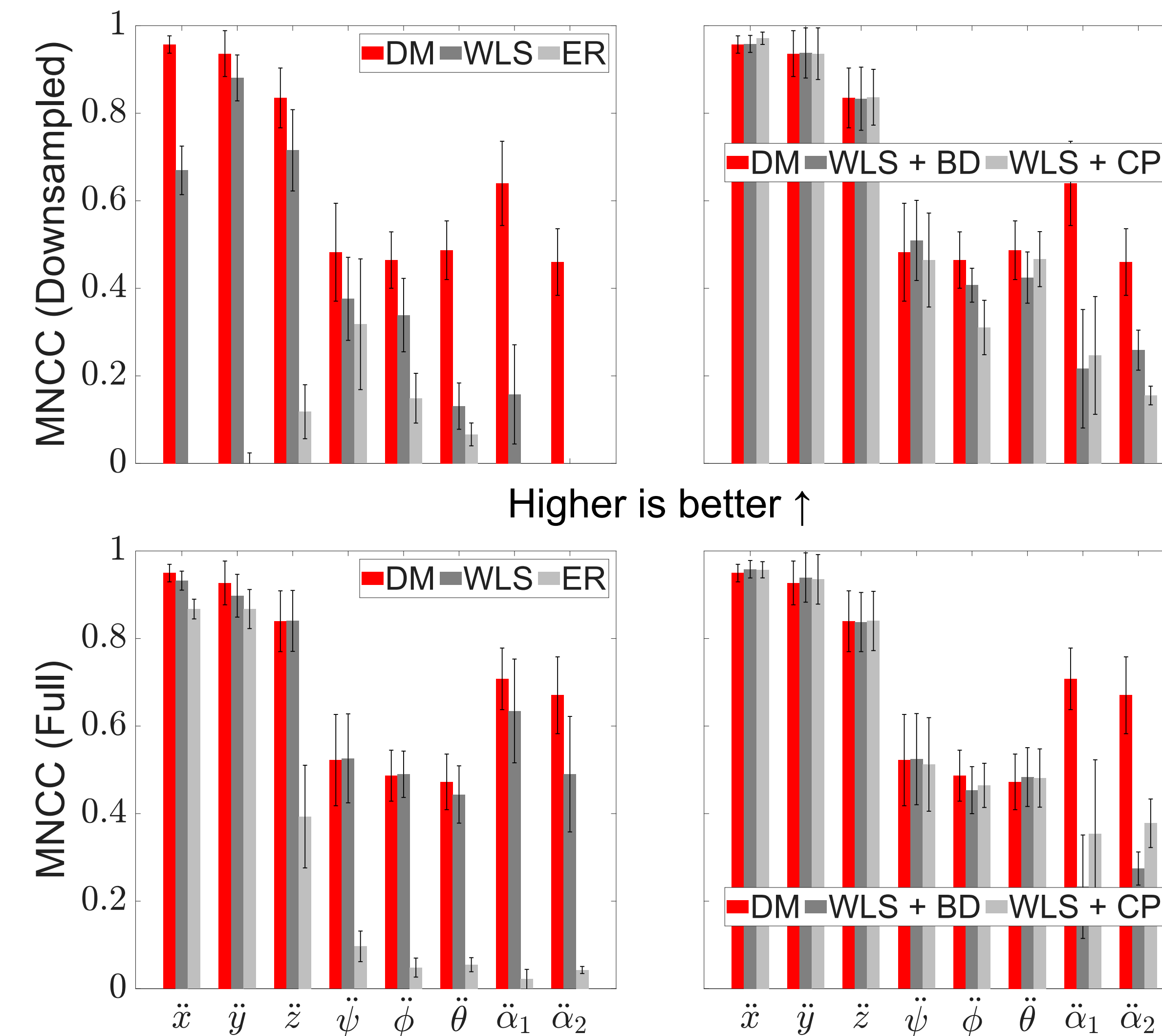
Evaluation: forward-dynamics acceleration prediction

Six training and six test trajectories with non-harmonic sinusoidal excitation in the flat outputs (~ 35 s each); motion capture and onboard sensing at 100 Hz.

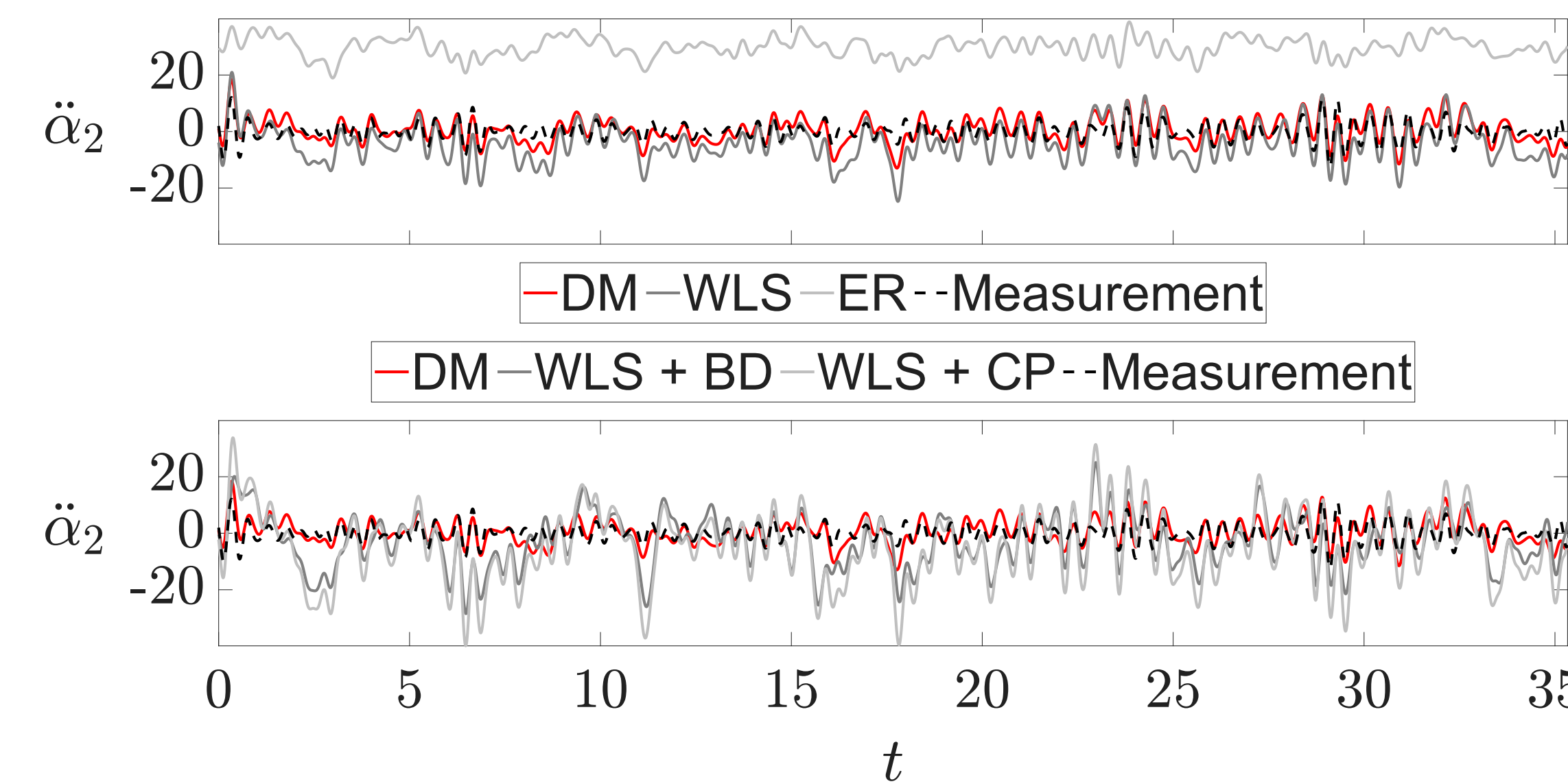


Experimental hardware and representative aggressive-maneuver prediction. Black dashed: ground truth; red: DM; blue: WLS.

Training data: ~ 20 downsampled / $\sim 21,000$ full samples.



Largest gains on pendulum joint accelerations. DM remains close to full-data performance after downsampling.

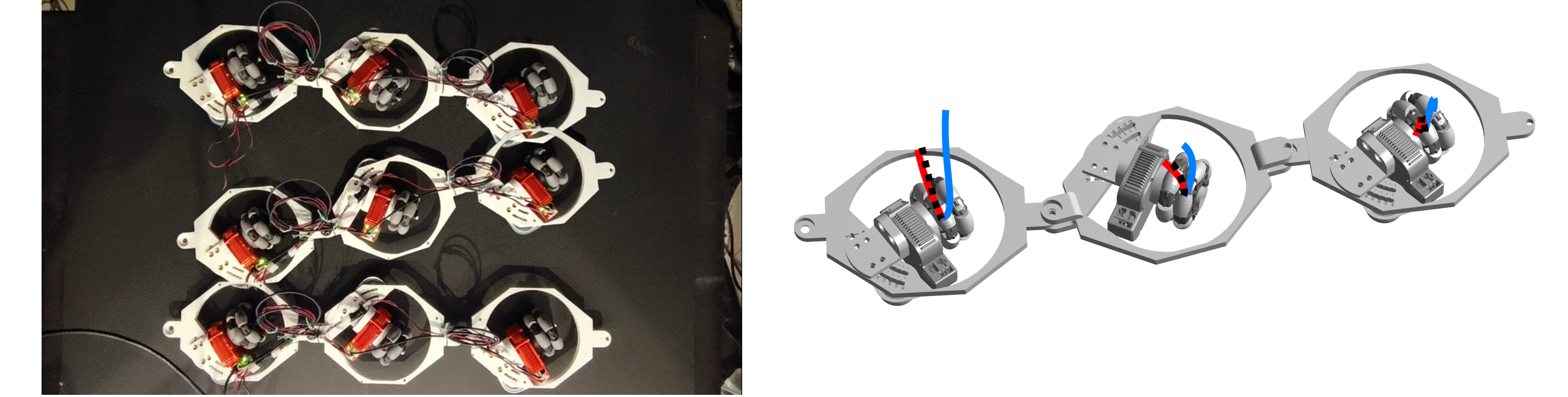


Full-data prediction α_2 . DM better tracks pendulum motion.

04 Drag-dominated: three-link LandSalp

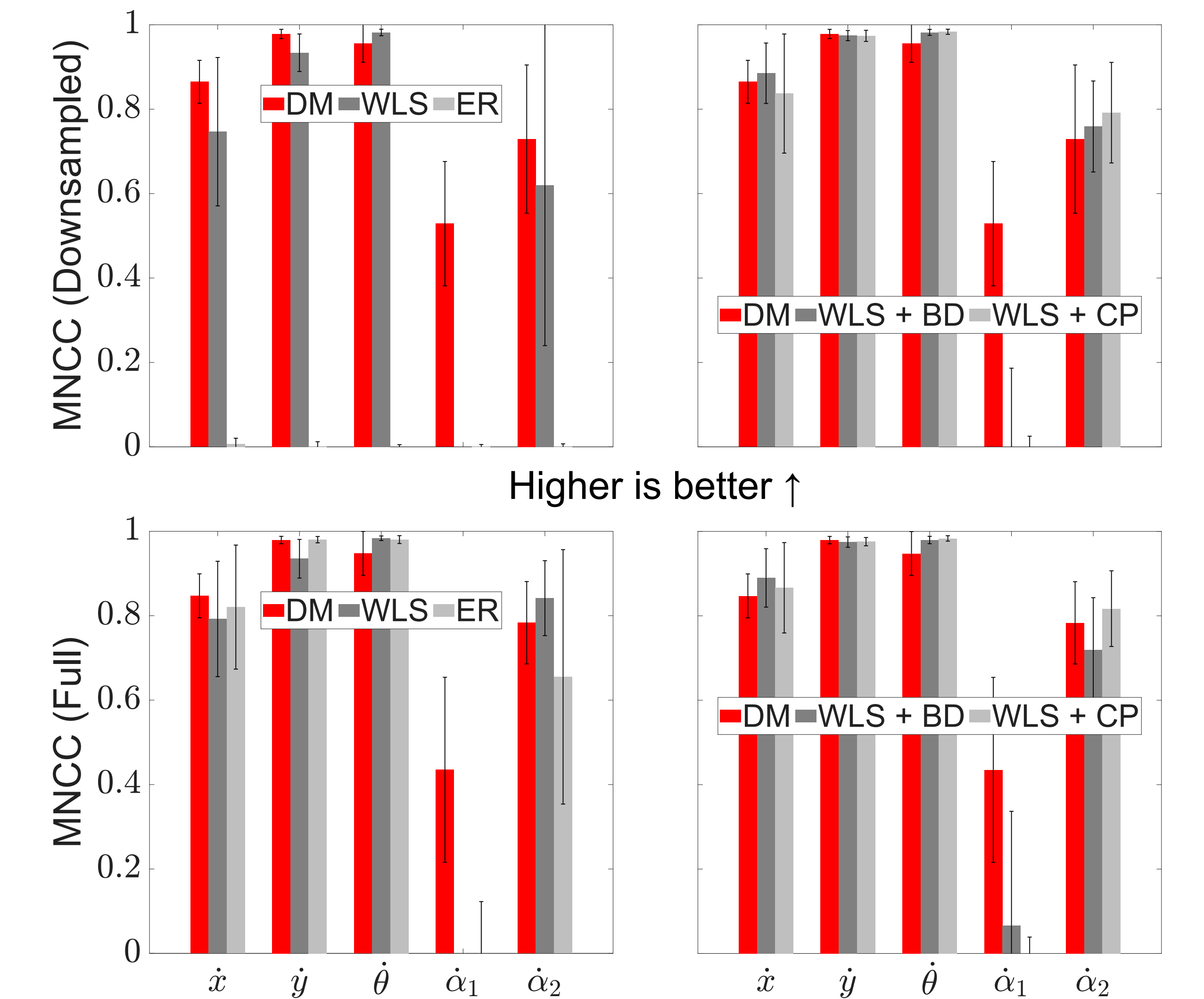
Evaluation: forward-dynamics velocity prediction

Eight excitation segments (~ 3 min) for training; directional and turning gaits for testing; motion capture and actuator force sensing.

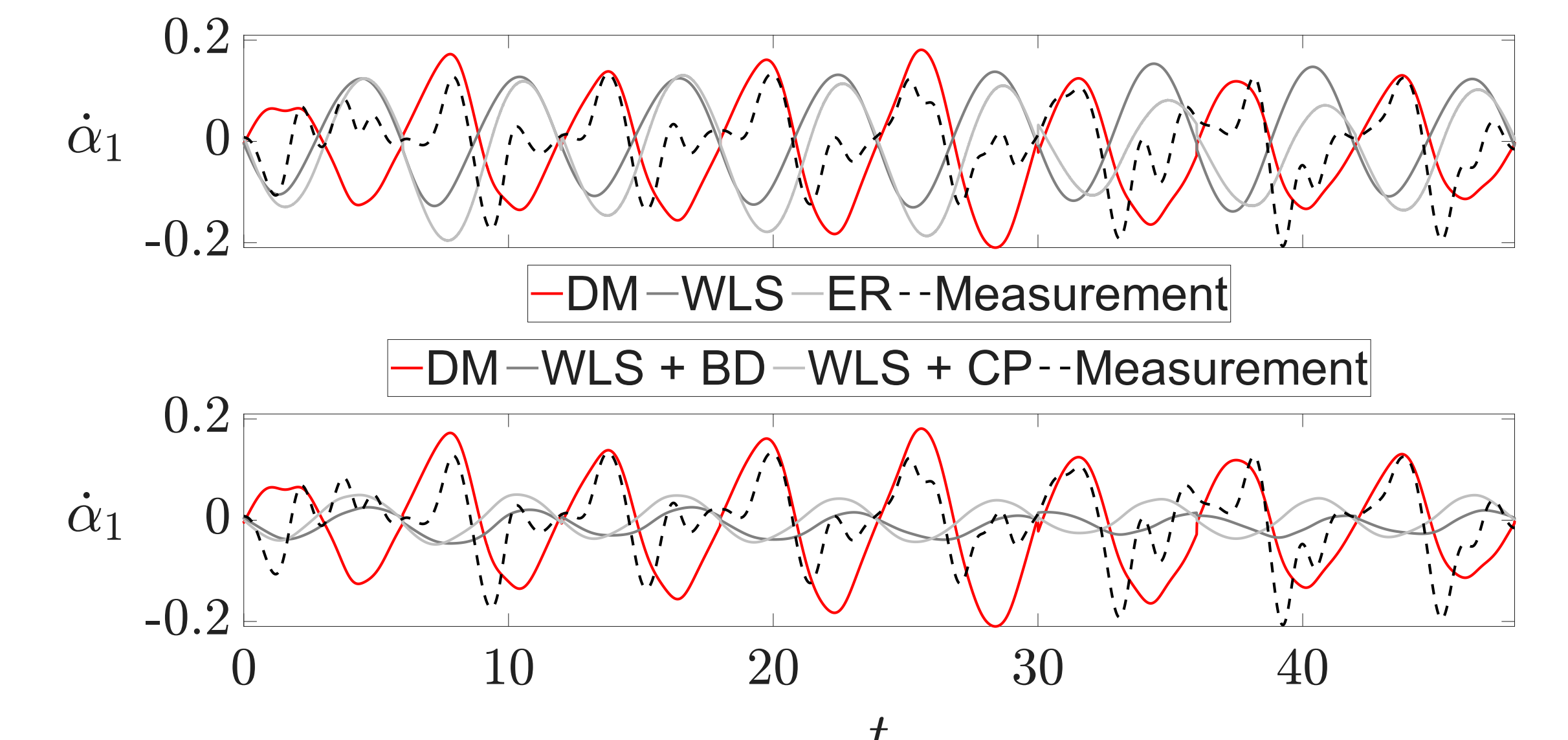


Experimental hardware and representative aggressive-maneuver prediction. Black dashed: ground truth; red: DM; blue: WLS.

Training data: ~ 40 downsampled / $\sim 38,000$ full samples.



Largest gains on passive-joint velocities. DM remains close to full-data performance after downsampling.



Full-data prediction α_1 . DM better captures passive-joint motion and phase.

DM: dual metric; **WLS:** error-covariance-weighted least squares; **ER:** energy-based residual; **BD / CP:** Bregman / constant-pullback regularization.

MNCC: maximum normalized cross-correlation ($\uparrow$); error bars: test-set standard deviation.

All fits enforce physical consistency.

Solve times (DM / WLS): Crazyflie 12.4 / 6.5 s; LandSalp 19.5 / 3.9 s. MOSEK, Intel Core i5-14400F.